

AI-Enabled Smart Campus With Wireless Sensor Network

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Abstract—Managing electricity in educational institutions is a critical environmental and financial challenge. Campus energy monitoring based on Wireless Sensor Networks (WSNs) commonly relies on static thresholds that ignore context, leading to missed wastage and alert fatigue. This paper presents an AI-enabled smart campus framework that couples a WSN sensing layer with an AI analytics layer for context-aware energy anomaly detection. The framework has five layers: sensing, communication, gateway, AI analytics, and application/alerting. The WSN layer is specified at the component level but not yet deployed. The AI analytics layer was therefore developed and evaluated on the UCI Appliances Energy Prediction dataset as a structural proxy for WSN telemetry. The analytics layer combines cyclical temporal features, rolling statistics, and strict autoregressive lags with an LSTM forecaster, which achieved the highest accuracy on the held-out test set ($R^2 = 0.575$) and reduced MAE by 31% over a persistence baseline. A forecast-residual detector with hour-dependent upper thresholds achieved a mean F1-score of 0.90 on synthetic injected anomalies over 30 randomized trials and outperformed the static, global- σ , and Isolation Forest baselines. The analytics layer is served through a stateful FastAPI backend with Redis and SQLite. Results are from a proxy study and indicate feasibility; validation on real WSN data from classrooms is future work.

Index Terms—Smart Campus, Wireless Sensor Network (WSN), Artificial Intelligence, Energy Monitoring, LSTM, Anomaly Detection

I. INTRODUCTION

Universities manage large infrastructures, and energy use and sustainability remain central open challenges in smart campus research [1]. A significant share of electrical energy can be wasted in classrooms where lighting, HVAC, and projectors stay on in unoccupied spaces. A smart campus addresses this by pairing a Wireless Sensor Network (WSN), which measures consumption and environment at the room level, with an AI layer that interprets the telemetry in context.

Research Gap: Many IoT-based monitoring and control systems act on fixed lower and upper limits of sensed variables [2]. A fixed energy alert (for example, a warning above 500 Wh per interval, used here only as an illustration) ignores time of day: a high reading is expected during a lecture but anomalous overnight, which causes alert fatigue. This work proposes an AI-enabled smart campus framework in which a WSN supplies telemetry and a forecasting-based analytics layer applies dynamic, hour-dependent residual thresholds.

Contributions:

- 1) A layered AI-enabled smart campus architecture with a WSN sensing layer, specified at the component level.
- 2) A context-aware data pipeline for WSN telemetry, structured to prevent autoregressive leakage.
- 3) A comparative, ablation-backed evaluation of forecasting models using time-series validation on a proxy dataset.
- 4) A forecast-residual anomaly detector with hour-dependent thresholds, benchmarked against static, global- σ , and Isolation Forest baselines on synthetic anomalies.
- 5) A stateful serving design bridging MQTT telemetry to HTTP-based LSTM inference.

Scope: No physical WSN has been deployed yet. The WSN layer is a design specification, and all quantitative results come from the proxy dataset (Section IV onward).

II. RELATED WORK

Data-driven building energy prediction has been widely reviewed [3], with methods ranging from statistical models to tree ensembles (Random Forest, XGBoost) and recurrent networks such as LSTM. For anomaly detection, the literature contrasts unsupervised methods such as Isolation Forest and autoencoders with forecasting-based residual approaches [4], which flag deviations from a predicted baseline without labeled anomalies. Prior IoT-based smart campus studies mainly address basic telemetry collection and rule-based automation. Compared with these, this paper adds an AI analytics layer designed to sit on top of a campus WSN.

III. PROPOSED AI-ENABLED SMART CAMPUS ARCHITECTURE

A. Layered Design

The system architecture, summarized in Table I, outlines a pipeline from edge telemetry to predictive alert generation.

TABLE I
LAYERED SMART CAMPUS ARCHITECTURE

Layer	Components	Role	Status
Sensing (WSN)	ESP8266, PZEM-004T, DHT11	Room telemetry (10 min)	Designed
Comm.	Wi-Fi (MQTT protocol)	Node-to-gateway transport	Designed
Gateway	Raspberry Pi 4	MQTT-to-HTTP Bridge	Designed
AI Analytics	LSTM, FastAPI, Redis	Prediction & anomaly logic	Proxy-Eval.
Application	SQLite, severity flags	Storage & severity flags	Implemented

This architecture utilizes ESP8266 nodes communicating via MQTT over a Wi-Fi star topology. Given the 10-minute sampling interval, the data payload is highly lightweight, making standard Wi-Fi sufficient and consistent with options surveyed in recent smart building designs [5]. The gateway only bridges MQTT messages to HTTP requests; the analytics services run on a separate server. Figure 1 illustrates the logical data flow.

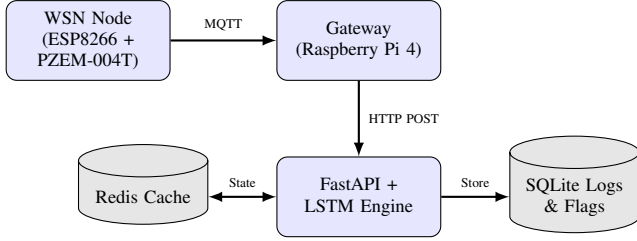


Fig. 1. System architecture showing the flow from WSN edge telemetry through the MQTT-to-HTTP gateway to the stateful backend analytics.

B. Telemetry Format

Each node posts a JSON payload with node ID, timestamp, energy (Wh per 10-minute interval), temperature, and humidity. All energy values use Wh per interval throughout the paper.

IV. PROXY DATASET AND PREPROCESSING

Because classroom WSN data is not yet available, the analytics layer was developed on the UCI Appliances Energy Prediction dataset [6]: 19,735 intervals at 10-minute resolution over 137 days.

Curation and leakage prevention: The target-correlated “lights” variable and two random-noise variables ($rv1$, $rv2$) were removed. The target is right-skewed, so a $\log(1 + y)$ transform was used for training. All metrics are reported on the back-transformed scale (Wh).

Features (Table II): The final input space contains 27 features. All lag and rolling features use $\text{shift}(1)$ so only past telemetry is used. Cyclical encoding is applied via $Hour_{sin} = \sin(2\pi \cdot \text{hour}/24)$ and $Hour_{cos} = \cos(2\pi \cdot \text{hour}/24)$.

TABLE II
FEATURE SET (27 FEATURES)

Category	Features
Environmental	T1-T9, RH_1-RH_9, T_out
Temporal	$Hour_{sin}$, $Hour_{cos}$, Is_Weekend
Autoregressive	Lag_1 (10 min), Lag_6 (1 h), Lag_{144} (24 h)
Rolling	$Mean_{1h}$, Std_{1h}

Heteroskedasticity: Figure 2 shows the per-hour residual standard deviation (σ_h), which peaks in the evening. This structurally motivates thresholds that vary by hour.

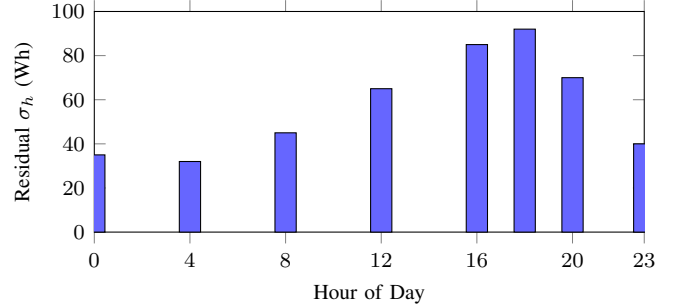


Fig. 2. Per-hour residual standard deviation (σ_h) computed from the validation set.

V. FORECASTING MODELS AND RESULTS

The task is one-step-ahead forecasting of y_{t+1} . Baselines include persistence and seasonal naive (same hour on the previous day).

Training Details: Hyperparameters were tuned on the validation set, with final evaluation performed exclusively on the 15% held-out test set (≈ 20.5 days). A fixed random seed (42) was utilized for reproducibility.

- **Random Forest:** 200 estimators, max depth 12.
- **XGBoost:** 100 estimators, learning rate 0.05, max depth 6.
- **LSTM:** Two layers (64, 32 units), dropout 0.2, 12-step (2 h) lookback. Trained via Adam optimizer, 50 epochs, batch size 32, with an early stopping patience of 5.

An ablation study on the validation set (Table III) indicated that autoregressive and cyclical features contributed to model performance, while rolling statistics contributed only marginally.

TABLE III
ABLATION (XGBOOST BASELINE ON VALIDATION SET)

Configuration	R ²
Full feature set (27 features)	0.545
Without cyclical temporal features	0.512
Without rolling statistics	0.538
Without autoregressive lags	0.390

TABLE IV
MODEL PERFORMANCE (15% HELD-OUT TEST SET)

Model	MAE (Wh)	RMSE (Wh)	R ² (95% CI)
Naive (persistence)	34.65	88.88	0.055 [0.01, 0.09]
Seasonal naive	29.10	72.30	0.285 [0.23, 0.33]
Linear regression	31.50	70.40	0.355 [0.32, 0.38]
Random Forest	27.20	64.10	0.495 [0.47, 0.51]
XGBoost	25.80	61.50	0.530 [0.51, 0.55]
LSTM	23.73	59.39	0.575 [0.56, 0.59]

As detailed in Table IV, the block-bootstrap confidence intervals of R^2 indicate that the LSTM achieves the best accuracy among the evaluated models, with a 31% MAE reduction over persistence. A Diebold-Mariano test indicated that the LSTM’s forecast errors were significantly lower than those of XGBoost ($p < 0.01$).

Classroom-Realistic Subset: To address domain shift, a secondary experiment trained the LSTM utilizing only sensors expected on a classroom edge-node (T_1 , RH_1 , time, lags, rolling stats). Performance dropped modestly ($R^2 = 0.542$), which indicates that useful accuracy may be retained with a restricted WSN feature set.

VI. ANOMALY DETECTION

Detector: For each interval, the residual is $r_t = y_t - \hat{y}_t$. Using residuals strictly from the validation period, the per-hour mean (μ_h) and standard deviation (σ_h) are computed. A one-sided threshold flags anomalous intervals: $r_t > \mu_h + 3\sigma_h$.

Injection Protocol: In the absence of labeled ground truth, 50 synthetic events were injected into the test set (comprising ~ 3000 steps). Injections included 15 subtle spikes ($+50$ – 100 Wh, strictly placed during nocturnal low-variance hours), 20 large spikes ($3 \times$ local standard deviation, lasting 1 interval), and 15 sustained overnight loads ($+200$ Wh for 18 intervals).

Anomaly-Gated State Update: To prevent sustained anomalies from contaminating autoregressive lag features, the system implements a stateful override: if y_t is flagged, the predicted $\hat{y}_t$ is written to the lag cache. To prevent the model from locking out real level-shifts (e.g., timetable changes), this override is strictly capped at 6 consecutive intervals (1 hour). Rolling statistics are computed on these corrected values. The Isolation Forest baseline utilized four features (Lag_1 , Lag_6 , Hour, Energy) with a 0.05 contamination rate.

TABLE V
ANOMALY BENCHMARK (EVENT-LEVEL, 30 RANDOMIZED TRIALS)

Method	Precision	Recall	F1	FA/day
Tuned Static (99th Pct)	0.65	0.52	0.58	1.44
Global $\mu + 3\sigma$	0.72	0.68	0.70	1.80
Isolation Forest	0.85	0.78	0.81	7.20
Hourly $\mu_h + 3\sigma_h$	$0.92 \pm .02$	$0.88 \pm .03$	$0.90 \pm .02$	$0.24 \pm .05$

Table V shows that, on the proxy data, the hour-dependent detector achieved the highest F1-score with the lowest false-

alarm rate (calculated over 144 daily intervals). The Isolation Forest alarm rate is governed by its contamination parameter ($0.05 \times 144 = 7.2$ per day) and therefore reflects that setting rather than a tuned operating point. Detection recall varied by type: large spikes (95%), sustained loads (93%, average delay of 1 interval), and nocturnal subtle spikes (73%). Figure 3 illustrates a flagged event and a missed subtle anomaly.

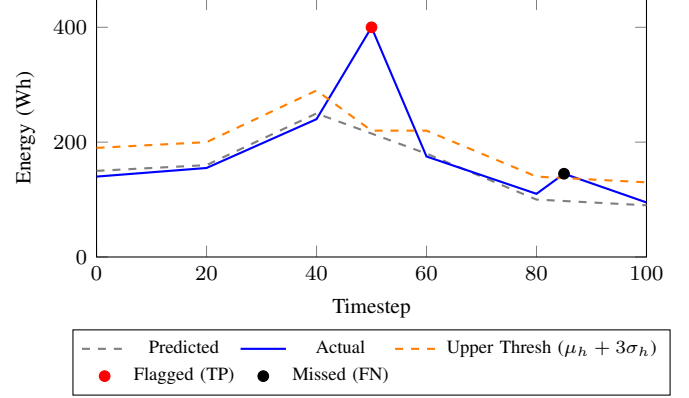


Fig. 3. Schematic illustration of predicted vs. actual consumption, showing a flagged spike (true positive) and a missed subtle anomaly below the threshold (false negative).

VII. BACKEND IMPLEMENTATION

The analytics layer is exposed through an asynchronous FastAPI service. The endpoint `/api/sensor-data` accepts JSON payloads routed by the gateway, runs inference, updates the anomaly-gated cache, and returns a severity flag.

Stateful Serving: Because LSTM features rely on Lag_{144} and 12 sequential inputs, the Redis cache stores the most recent 144 intervals (24 hours) per node. During a cold start, a new node relies on a seasonal naive fallback for 24 hours. Missing packets are forward-filled. End-to-end API latency averages 42 ms per packet (including 18 ms for LSTM inference, batch size 1).

VIII. LIMITATIONS AND FUTURE WORK

- 1) **No deployed WSN:** The sensing and communication layers are design specifications pending hardware validation.
- 2) **Domain shift:** A physical classroom WSN will possess a smaller feature set (DHT11, PZEM-004T), requiring empirical retraining.
- 3) **Synthetic anomalies:** Detection results represent mathematically simulated constraints rather than empirical wastage events.
- 4) **Evaluation scope:** Results come from a single residential dataset, a single injection design, and a single training seed. Subtle spikes were placed only in nocturnal low-variance hours, which favors hour-dependent thresholds.

Future work: Deploy ESP8266 + PZEM-004T + DHT11 nodes with a Raspberry Pi 4 gateway in classrooms, collect

empirical data for 14 days, retrain the LSTM, add a class-timetable feature, and benchmark against real wastage events.

IX. CONCLUSION

This proxy-dataset study evaluates the software feasibility of a prediction-based anomaly detection framework for WSN energy monitoring. Time-series evaluation indicates that an LSTM forecaster ($R^2 = 0.575$) combined with hour-dependent statistical thresholds ($\mu_h + 3\sigma_h$) is a context-aware alternative to static alerting. With an F1-score of 0.90 against synthetic anomalies and a low false-alarm rate on the proxy data (0.24 FA/day), the stateful API architecture offers a basis for future smart-campus deployments, pending validation on real WSN data.

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CODE AND DATA AVAILABILITY

The backend architecture will be open-sourced at <https://github.com/codedbyadiiii/IntelligentClassroomEnergy>. The UCI dataset is publicly available [6].

REFERENCES

- [1] N. Chagnon-Lessard et al., “Smart campuses: Extensive review of the last decade of research and current challenges,” *IEEE Access*, vol. 9, pp. 124200–124234, 2021. <https://doi.org/10.1109/ACCESS.2021.3109516>
- [2] P. Paudel, S. Kim, S. Park, and K.-H. Choi, “A context-aware IoT and deep-learning-based smart classroom for controlling demand and supply of power load,” *Electronics*, vol. 9, no. 6, Art. 1039, 2020. <https://doi.org/10.3390/electronics9061039>
- [3] H. Liu, J. Liang, Y. Liu, and H. Wu, “A review of data-driven building energy prediction,” *Buildings*, vol. 13, no. 2, Art. 532, 2023. <https://doi.org/10.3390/buildings13020532>
- [4] V. Chandola, A. Banerjee, and V. Kumar, “Anomaly detection: A survey,” *ACM Comput. Surv.*, vol. 41, no. 3, Art. 15, pp. 1–58, Jul. 2009. <https://doi.org/10.1145/1541880.1541882>
- [5] N. Naji, M. R. Abid, N. Krami, and D. Benhaddou, “Energy-aware wireless sensor networks for smart buildings: A review,” *J. Sens. Actuator Netw.*, vol. 10, no. 4, Art. 67, 2021. <https://doi.org/10.3390/jsan10040067>
- [6] L. M. Candanedo, V. Feldheim, and D. Deramaix, “Data driven prediction models of energy use of appliances in a low-energy house,” *Energy and Buildings*, vol. 140, pp. 81–97, 2017. <https://doi.org/10.1016/j.enbuild.2017.01.083>